Industry 4.0 Misinformation Analytics

1. Problem identification In this project, we frame online misinformation moderation as a **Quality 4.0 optimisation problem** inside a cyber-physical "social factory". Social media posts are treated as products moving through a production line, harmful misinformation is the defect, and human fact-checkers are the constrained inspection resource.

The operational goal is to decide, for each post i, whether to send it to manual review. For each post we estimate a risk score $p_i = P(\text{misinformation} \mid \text{features})$ and have an estimate of its expected engagement (reach). The optimisation objective is to **maximise harmful engagement intercepted**:

$$\text{Maximise } \sum_i x_i \cdot (p_i \times \text{engagement}_i)$$

where the **decision variable** is

- $x_i = 1$: post i is routed to fact-checkers
- $x_i = 0$: post i is left to the automated pipeline

subject to realistic Industry 4.0 style **constraints**:

- **Global review capacity** $\sum_i x_i \leq K$, where K is the maximum number of posts that can be manually reviewed per day.
- **Platform-level quotas (optional)** $\sum_{i \in \text{platform } j} x_i \leq K_j$, so that no single platform (e.g., Twitter) consumes the entire review capacity.
- **Reviewer time budget (optional)** $\sum_i x_i \cdot \text{token\_count}_i \leq T$, where `token_count` approximates reading/inspection time and T is the total reviewer time available per cycle.

This mirrors **supplier screening in Quality 4.0**: suppliers ≈ authors, lots ≈ posts, defect probability ≈ misinformation risk, and the reviewer workforce ≈ constrained inspection cell. The key Industry 4.0 decision is not only "is this post risky?" but **"given limited human capacity, which risky posts should be seen first so that we intercept the maximum harmful engagement?"**

2. Relevant data To support this optimisation problem, we constructed a dataset of **500 posts** from **Twitter, Facebook, Telegram, and Reddit** over 12 months, containing **31 raw variables + 4 engineered features** (posting hour, human-readable label, log-scaled followers/engagement). The target indicates misinformation vs legitimate content.

Features reflect multiple Industry 4.0 information streams:

1. **Platform, geography, and time**: Platform type, country (US, India, Germany, Brazil, UK), temporal patterns (month, weekday, hour) for risk patterns and capacity rules.
2. **Author traits and reach**: Follower count (log-scaled), verification status, engagement metrics (reactions, comments, shares) as **harm proxy** in the objective.
3. **Content and linguistic signals**: Text length, token count, readability, sentiment/toxicity scores, URL/mention/hashtag counts.
4. **Reliability and AI provenance**: Source domain reliability, synthetic text detection, **generator model signature**, embedding similarity to factual content, fact-check counts/verdicts.

Key findings: * **Misinformation prevalence**: ~54% (naïve baselines inadequate) * **Twitter** shows highest misinformation rates; platform–country hotspots emerge (Twitter–US, Telegram–India) * Clear **temporal patterns** (May/September spikes, Friday/Monday peaks) support adaptive staffing * Misinformation posts are **longer, more negative, linked to lower-reliability domains**; engagement distributions similar across classes

The dataset provides exactly what optimisation requires: misinformation probability p_i, engagement-based harm measure, and contextual variables for realistic Industry 4.0 constraints.

3. Approach to solve the problem Our solution follows a **three-stage pipeline**: descriptive analytics → predictive modelling → prescriptive optimisation.

a) Exploratory analytics We began with detailed **exploratory data analysis (EDA)** to understand where risk concentrates and to justify feature choices:

- Profiled class balance, platform-wise misinformation rates, and platform–country heatmaps.
- Analysed monthly, weekday, and hourly trends to identify **temporal "campaign-like" spikes**.
- Examined follower and engagement distributions (heavy-tailed reach, similar engagement for both classes).

- Compared content-level features (length, tokens, readability, sentiment, toxicity, URLs/mentions/hashtags) between misinformation and legitimate posts.

This EDA established that risk is **multi-factorial and diffuse**, and it highlighted the specific segments (platforms, time windows, author types) for scenario analysis in the optimisation stage.

b) Predictive modelling Next, we built a **machine-learning pipeline** to estimate $p_i = P(\text{misinformation} \mid \text{features})$ for each post:

- **Text representation**
  - TF–IDF bigrams (1–2-grams), limited to 500 features
  - TruncatedSVD (50 components) to obtain dense "topic" dimensions from the sparse TF–IDF matrix
- **Feature fusion**
  - Concatenated SVD text topics with scaled numeric metadata (followers, engagement, length, sentiment, toxicity, reliability, synthetic score, similarity, token_count)
  - Added one-hot encoded categorical variables (platform, country, month, weekday, hour, fact-check verdict, generator signature)
- **Models**
  - Class-balanced **logistic regression** (saga solver) as a linear baseline
  - Class-weighted **Random Forest** with 400 trees as a non-linear ensemble

Using a **70/30 stratified train–test split**, the Random Forest delivers:

- **Recall on misinformation** ≈ 83.8% – it correctly flags more than 5 out of 6 harmful posts.
- **Precision** ≈ 54% – about half of flagged posts are truly harmful.

ROC-AUC is low due to mis-calibrated probabilities, but the **ranking and class predictions** are strong enough for triage. Feature-importance analysis shows that no single feature dominates; instead, **token_count, sentiment, toxicity, engagement, domain reliability, and text topic components** collectively drive decisions. This matches the EDA conclusion that the signal is spread across many weak indicators.

c) Prescriptive optimisation and Industry 4.0 link Finally, we turn model outputs into **operational decisions** via a risk-weighted prioritisation policy. For each post:

- Compute expected harm $h_i = p_i \times \text{engagement}_i$
- Sort posts in descending order of h_i.
- For any review capacity K, send the top K posts to reviewers.

We evaluate this policy using a **coverage curve** that compares random review and risk-weighted review for different capacities (10, 20, 50, 100, 150 posts). The results show that:

- Reviewing only the **riskiest 30% of posts captures ~56% of total harmful engagement**, almost **2×** better than random sampling at the same capacity.
- Even at 10–20% capacity, the prioritisation consistently delivers around **1.8–1.9× more harmful engagement intercepted per reviewed post** compared to random review.

From an **Industry 4.0 perspective**, this closes the loop from analytics to action:

- Sensor-like data streams (post content and metadata) →
- Digital intelligence (predictive model estimating p_i) →
- Optimisation layer that **allocates scarce human "inspection" capacity** where it reduces risk the most.

This architecture can be extended into a **digital twin of the moderation process**, with adaptive thresholds based on temporal risk, continuous feedback from moderators, and fairness constraints across platforms and regions.

Repository: github.com/ALikesToCode/industry-4.0-gaabs-analysis